A

Pre-configured SaaS · Pillar A

Turnkey Intelligent Modules

High-efficiency, production-tested systems, ready for deployment.

A.1

Inventory & Distributed Warehousing Logic

Stock position per node, reservations and transfers.

Stock | Transfers | Reservations | Transfer

SKU	WH-A	WH-B	WH-C	Total
SKU-1042	128	64	0	192
SKU-2210	12	40	33	85
SKU-3087	220	0	18	238
SKU-4410	7	7	7	21
SKU-5002	54	90	31	175

● Synced 0.4 s ago — 5 SKUs · 3 nodes

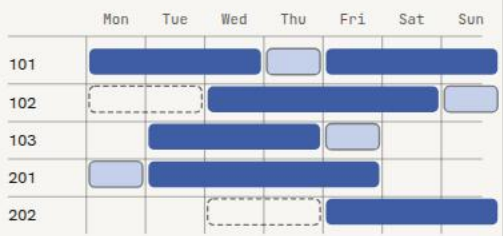
Topology	n-node, multi-warehouse
Sync	event-sourced, deterministic merge
Reservations	per-channel holds, auto-expiry
Offline	node-local ledger, reconcile on reconnect
Latency	< 100 ms in-site, sub-second across channels

A.2

Hospitality & Unit Management Engines

Units, stays, availability and turnover tasks.

Units | Housekeeping — This week



Occupied · Turnover · Blocked — no overlaps

Allocation	overlap-free by construction
Housekeeping	status transitions create tasks
Channels	booking channels kept in step
Rates	plans, seasons and restrictions
Handover	turnover opens at check-out

A.3

Unified Commerce & POS Ledger Sync

The transaction ledger: every entry, posted once.

Sync | Close — live

POS #A-10441

	24.00
	18.50
VAT	4.25
Total	**46.75**

Account	Dr	Cr
1100 Cash	46.75	·
4000 Sales	·	42.50
2200 VAT	·	4.25
Σ Dr = Σ Cr		46.75

● Balanced, posted once — idempotent

Model	double-entry, append-only
Idempotency	one key per transaction
Gateways	settlement matched automatically
Close	daily close with variance report
Audit	immutable history, exportable

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Inventory Units Ledger

Stock Transfers Reservations Transfer

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Stock position per node, reservations and transfers.

Fed by Warehouse scanners · 3PL & logistics

Topology	n-node, multi-warehouse
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```
type StockLevel {
  sku      string
  node     NodeId  // warehouse or store
  on_hand  int
  reserved int  // held for open orders
  version  lamport  // deterministic ordering
}
```

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A.2 Units

Units, stays, availability and turnover tasks. +

A.3 Ledger

The transaction ledger: every entry, posted once. +

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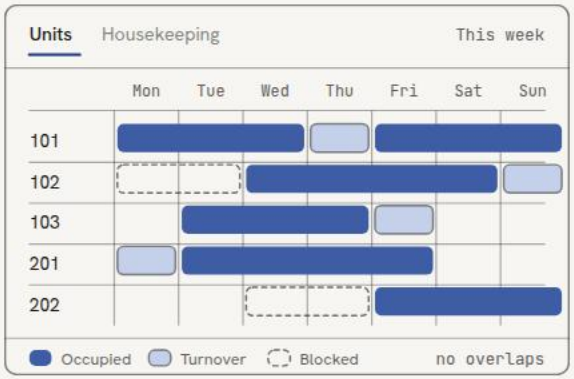

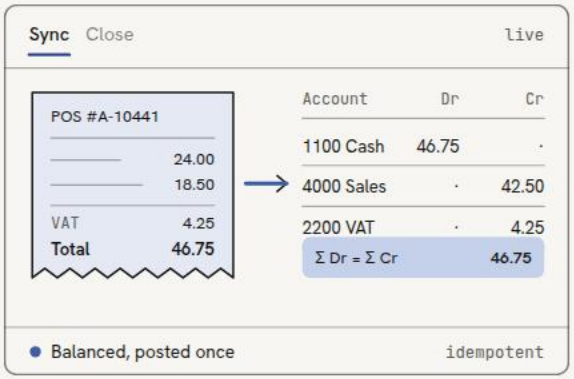

A.1 Inventory

Stock position per node, reservations and transfers.

Topology	n-node, multi-warehouse
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Reservations	per-channel holds, auto-expiry

A.2 Units

Units, stays, availability and turnover tasks.

Allocation	overlap-free by construction
Housekeeping	status transitions create tasks
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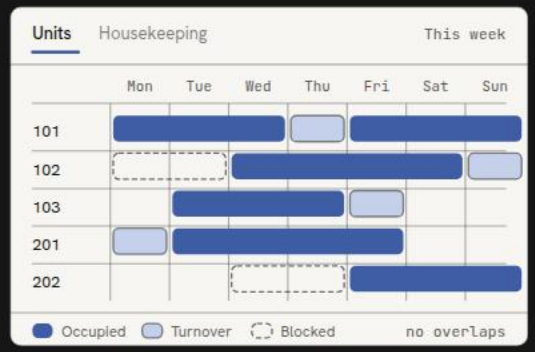

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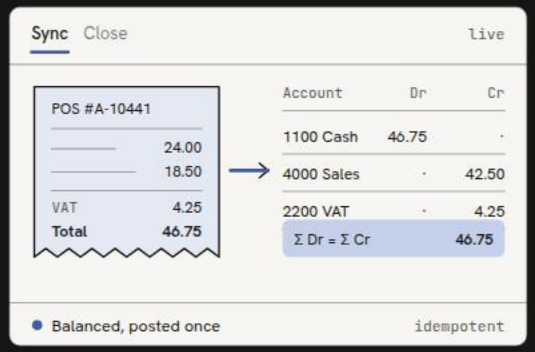

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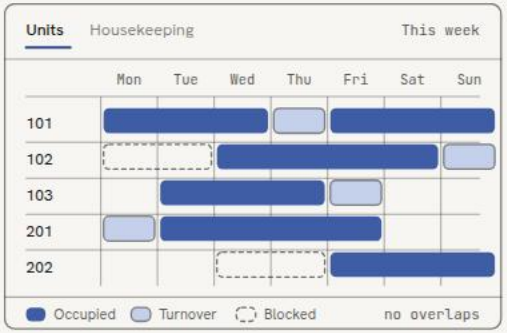

A.3 Ledger

Sync | Close | live

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Balanced, posted once · idempotent

	A.1 Inventory	A.2 Units	A.3 Ledger
Owns	Stock position per node, reservations and transfers.	Units, stays, availability and turnover tasks.	The transaction ledger: every entry, posted once.
Fed by	Warehouse scanners · 3PL & logistics	Booking channels · Property systems	Point of sale · Payment gateways

Parameters

Topology	n-node, multi-warehouse
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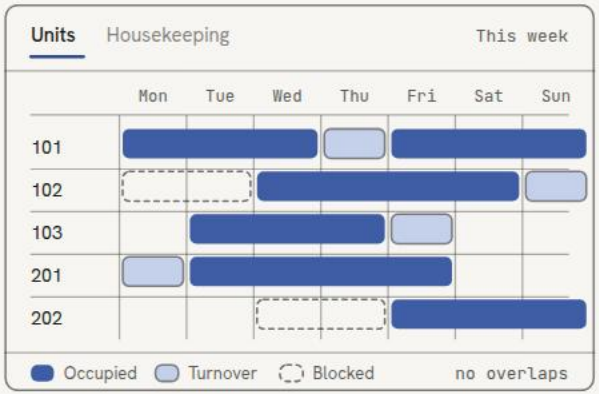

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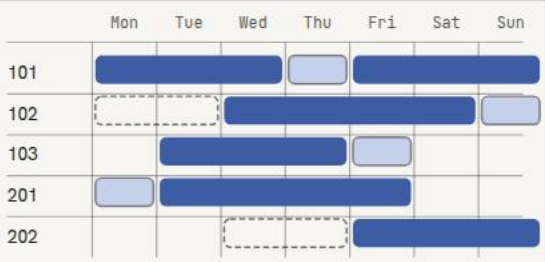

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